

Ryan Kim

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. Electrical Engineering & Computer Science • GPA: 3.82

Expected May 2029

Relevant Coursework: Data Structures & Algorithms, Machine Structures, Discrete Math, Circuits & Devices

EXPERIENCE

Teserac AI

Sunnyvale, CA

Datacenter operations startup. Four products across web, AI agents, mobile, and AR.

Software Engineer (Part-Time, 40 hrs/wk)

Mar. 2026 – May 2026, Aug. 2026 – Present

- **RayNeo X3 Pro AR App:** Built the hands-free technician app as its only engineer (Kotlin/Jetpack Compose): a wake word → VAD → on-prem Orin ASR → intent voice pipeline that resolves navigation intents to local deep links rather than the network, a WebRTC remote-expert mode with annotation overlay, on-device SAM 2 segmentation, and an offline-first cache with a retry-classifying outbox.
- **React Native Field App:** Built the platform's mobile field app end to end and shipped it to TestFlight: LiveKit voice with barge-in, Keycloak OIDC parity with web, a root error boundary with crash reporting, and signed-build release CI.
- **Auth & SSO Architecture Overhaul:** Overhauled email-domain SSO on Keycloak/OIDC, replacing DNS-based provider inference with a per-organization domain-route map and collapsing three user-provisioning paths into one endpoint; added session idle and absolute-age caps plus per-request ECDSA P-256 device proofs behind an off/monitor/enforce rollout.

Software Engineering Intern (Full-Time)

May 2026 – Aug. 2026

- **Graph API Architecture:** Re-architected the Digital Twin graph API, a browser-based single-line-diagram editor for datacenter power topology, into independently versioned planes. Wrote the topology and layout-intent planes, versioning operator layout by integer revision and SHA-256 hash so concurrent edits never overwrite each other.
- **Deterministic Layout & Canvas Performance:** Wrote the layout and grouping algorithms behind the editor: fixed-order synthetic ELK ports for reproducible component rows, a clear-then-regroup organize service that leaves no asset unclassified, and a structural topology fingerprint that keeps live power polling from forcing a relayout, rendering large diagrams roughly 3× faster.
- **Async Delete Pipeline & Stuck-Job Reaper:** Reworked facility cascade deletes after added telemetry pushed a single purge past millions of database operations, moving it onto a polled async job that chunks work into 2,000-op batches so deletes no longer time out. Wrote the reaper behind it, with a stale threshold above the task's hard limit so a running purge is never dispatched twice.

PROJECTS

AI Marketing Data Pipeline

Dec. 2025 – May. 2026

- **Multimodal Ingestion & Script Generation:** Deployed an end-to-end AI pipeline that turns raw roofing job photos and video into short-form marketing scripts with aligned timestamps, using FFmpeg and Deepgram speech-to-text for structured multimodal LLM input.
- **Commercial SaaS Revenue:** Productized it into a client-facing service generating \$650 in monthly recurring revenue.

SKILLS

Languages: Python, TypeScript, Kotlin, Java, SQL, JavaScript, Bash

Backend: Django REST Framework, FastAPI, LangGraph, Celery, PostgreSQL, TimescaleDB, Redis

Frontend: React, React Native, React Flow / ELK.js, TanStack Query, Jetpack Compose, Tailwind

Platform & Infra: Keycloak / OIDC, AWS S3, LiveKit + WebRTC, NVIDIA Orin, SAM 2, GitHub Actions